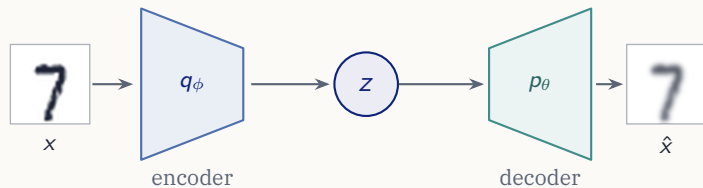


Generating New Data

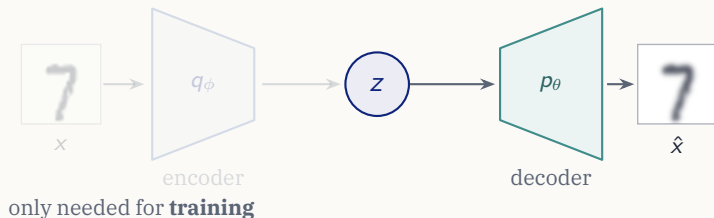
Training is over. To make *new* images, we only need half of the network.



While training, all three stages run: **encode**, **sample**, **decode**.

Generating New Data

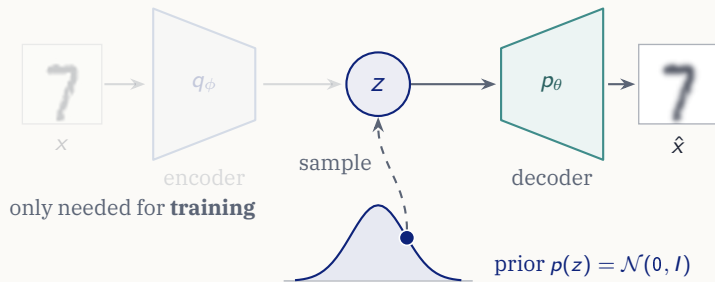
Training is over. To make *new* images, we only need half of the network.



To generate there is no input image to encode, so take z **straight from the prior**.

Generating New Data

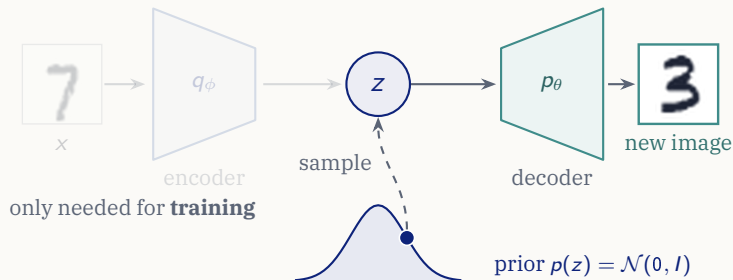
Training is over. To make *new* images, we only need half of the network.



To generate there is no input image to encode, so take z **straight from the prior**.

Generating New Data

Training is over. To make *new* images, we only need half of the network.



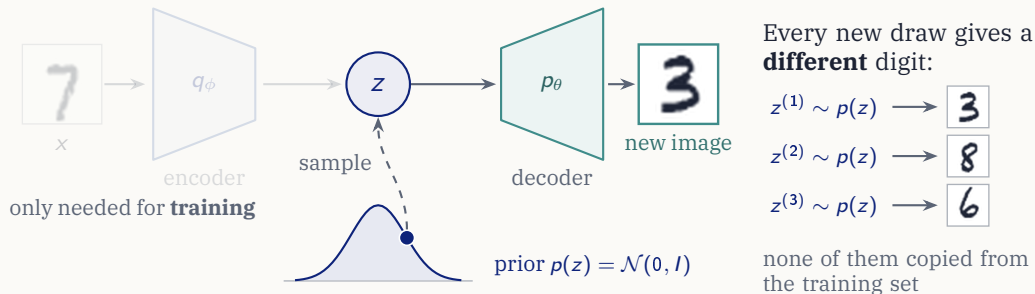
Every new draw gives a **different** digit:

$$\begin{aligned} z^{(1)} \sim p(z) &\longrightarrow \boxed{3} \\ z^{(2)} \sim p(z) &\longrightarrow \boxed{8} \\ z^{(3)} \sim p(z) &\longrightarrow \boxed{6} \end{aligned}$$

none of them copied from the training set

Generating New Data

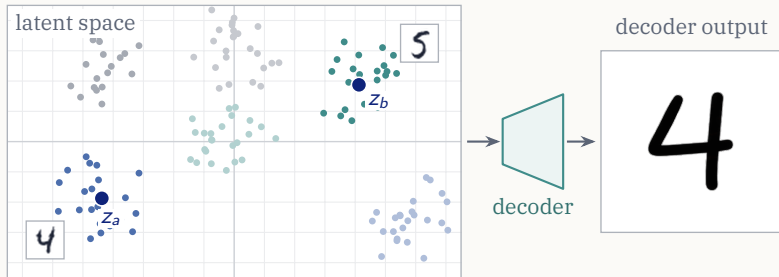
Training is over. To make *new* images, we only need half of the network.



The KL term trained every encoding to sit inside $\mathcal{N}(0, I)$, so a random draw lands where the decoder already knows what to draw. **Learning to generate, not just compress.**

Walking Through Latent Space

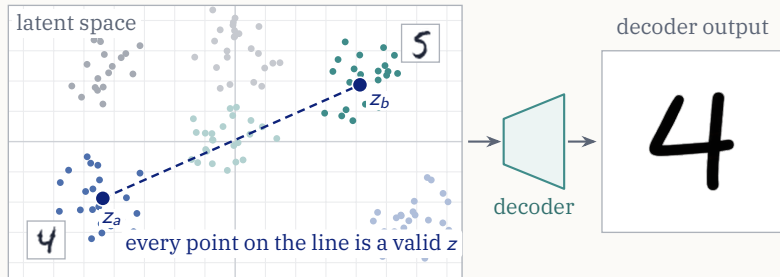
One sample gives one image. What if we don't jump, but **walk**?



At z_a , the decoder draws a **4**.

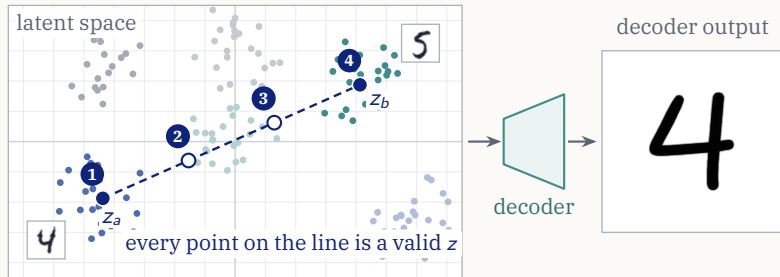
Walking Through Latent Space

One sample gives one image. What if we don't jump, but **walk**?



Walking Through Latent Space

One sample gives one image. What if we don't jump, but **walk**?



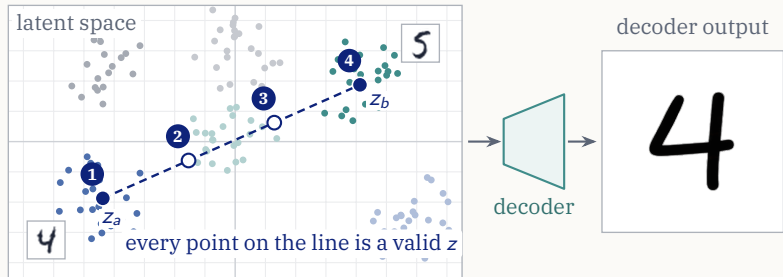
At z_a , the decoder draws a **4**.

Now slide z along the line to $z_b \dots$



Walking Through Latent Space

One sample gives one image. What if we don't jump, but **walk**?

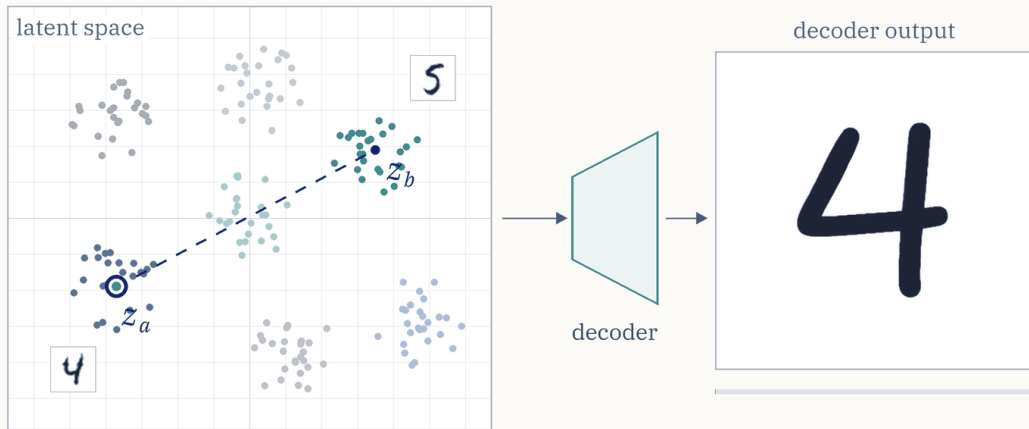


Every step
decodes to a
real-looking
digit.

Remember the
gaps from Part 1?
The KL term
filled them.



Walking Through Latent Space



Conditional VAE: Choosing What Comes Out

✗ **Vanilla VAE: we can't pick the digit**



Conditional VAE: Choosing What Comes Out

✗ Vanilla VAE: we can't pick the digit



✓ Conditional VAE (CVAE)

give the label y to **both** halves

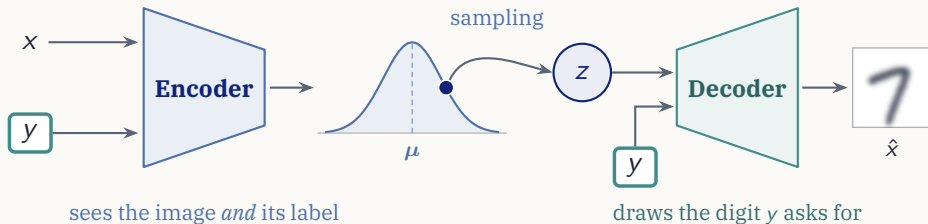
Conditional VAE: Choosing What Comes Out

✗ Vanilla VAE: we can't pick the digit



✓ Conditional VAE (CVAE)

give the label y to **both** halves



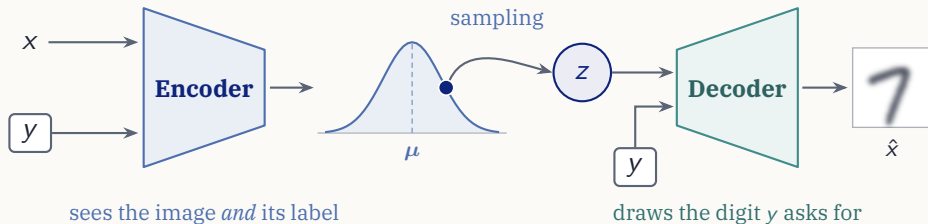
Conditional VAE: Choosing What Comes Out

✗ Vanilla VAE: we can't pick the digit



✓ Conditional VAE (CVAE)

give the label y to **both** halves



The label y says **which** digit.
So z only stores **how** it is written.

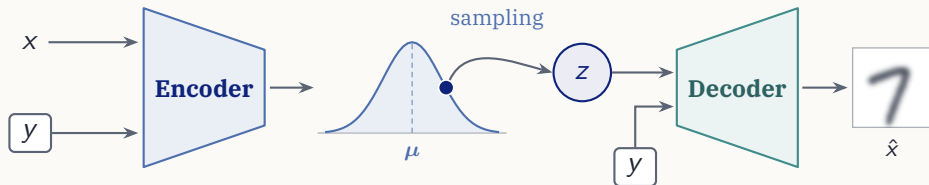
Conditional VAE: Choosing What Comes Out

✗ Vanilla VAE: we can't pick the digit



✓ Conditional VAE (CVAE)

give the label y to **both** halves



sees the image *and* its label

draws the digit y asks for

The label y says **which** digit.
So z only stores **how** it is written.

$y = 7$: **always a 7**,
only the *style* varies.



Where VAEs Are Used



Anomaly Detection

bad reconstruction = something unusual



Drug & Molecule Design

sample z to get new molecules

Where VAEs Are Used



Anomaly Detection

bad reconstruction = something unusual



Drug & Molecule Design

sample z to get new molecules



Recommendation

a latent “taste” for every user

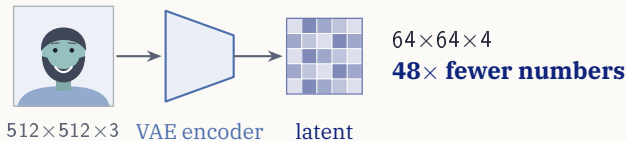


Data Augmentation

decode extra training examples

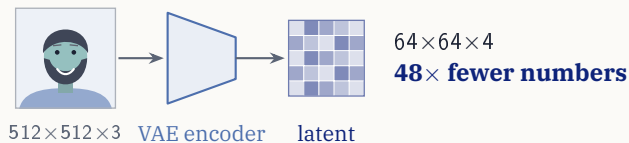
How Diffusion Models Use a VAE

1 Compress

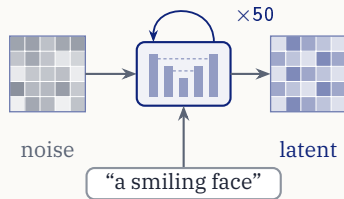


How Diffusion Models Use a VAE

1 Compress

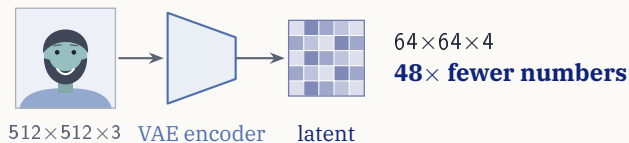


2 Generate

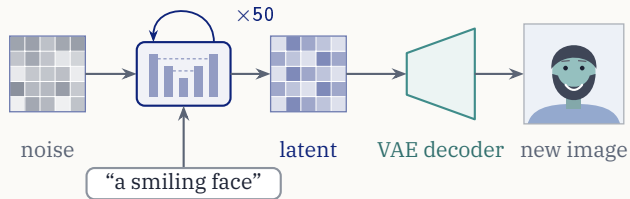


How Diffusion Models Use a VAE

1 Compress



2 Generate

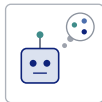


What's Next for VAEs



Video generation

Compressing space *and* time, so AI can make whole clips.



Robots that imagine

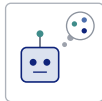
Practising inside a learned latent world before acting.

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Practising inside a learned latent world before acting.



New materials

Sampling new crystals for better batteries and solar cells.



Reading the brain

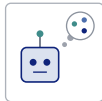
Turning brain scans back into the images a person saw.

What's Next for VAEs



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Compressing space *and* time, so AI can make whole clips.



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Practising inside a learned latent world before acting.



New materials

Sampling new crystals for better batteries and solar cells.



Reading the brain

Turning brain scans back into the images a person saw.

compress → **smooth** → **sample**: the same recipe, aimed at new problems.

Thank You